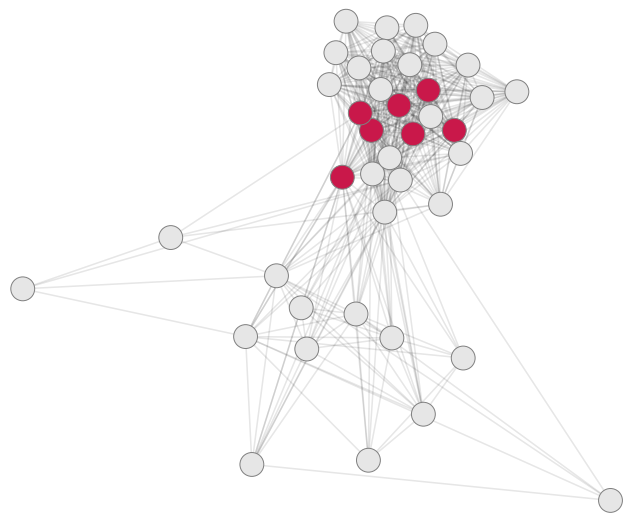
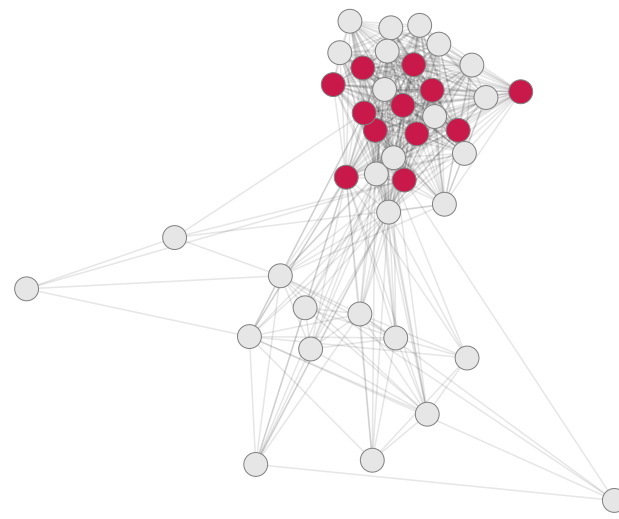


Spreading of resistance across the relatedness networks (cumulative; nodes light up and never revert)

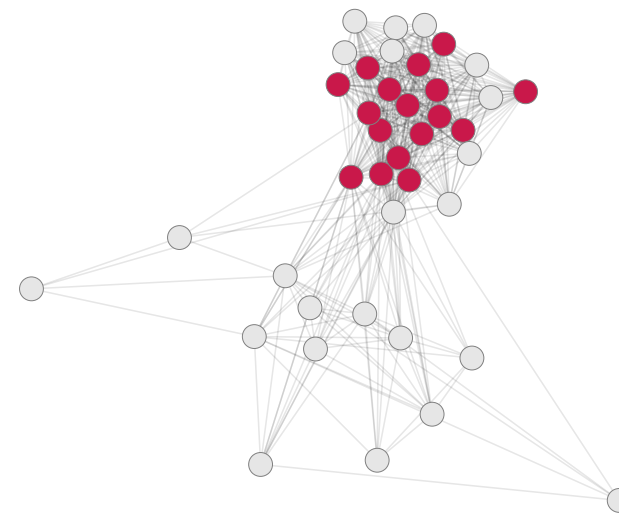
K. pneumoniae — 2009 (7 R)



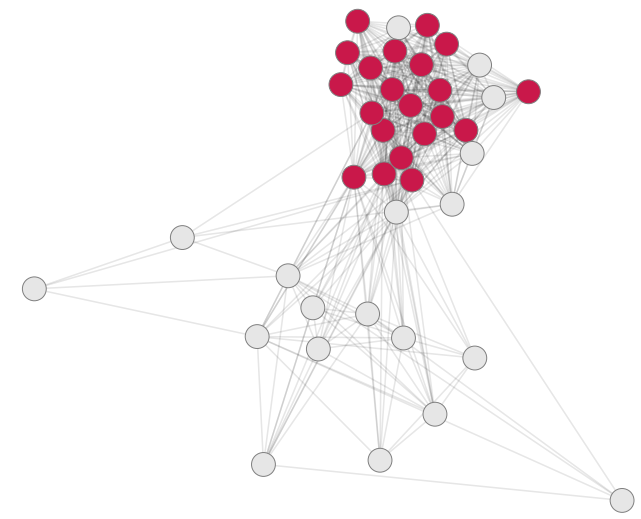
K. pneumoniae — 2014 (12 R)



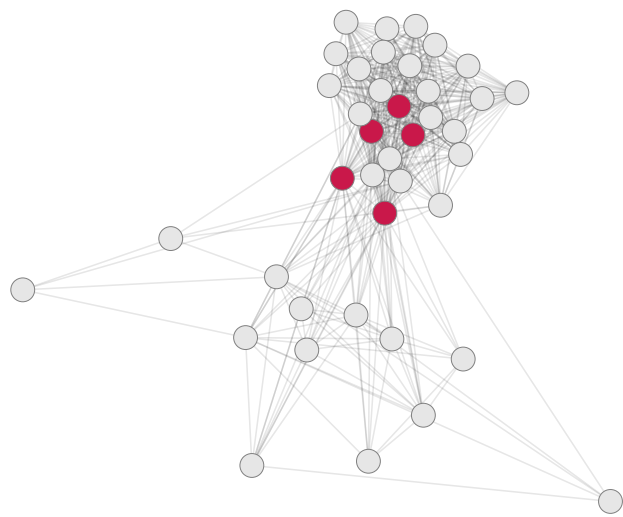
K. pneumoniae — 2019 (17 R)



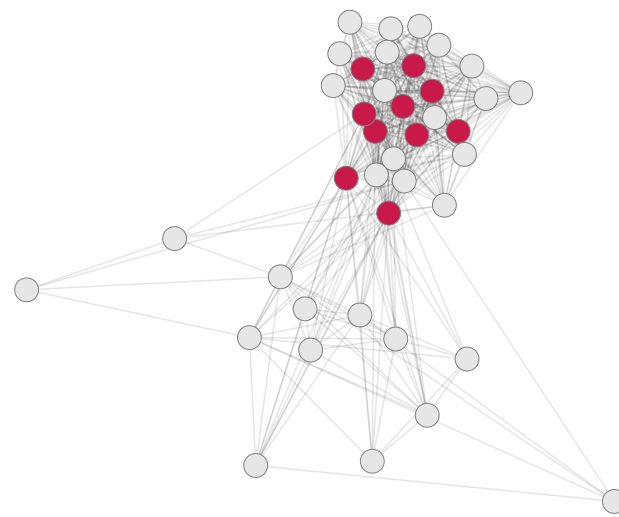
K. pneumoniae — 2024 (21 R)



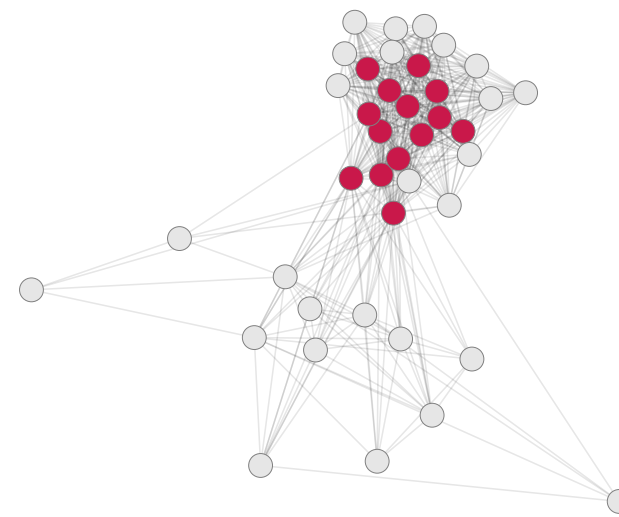
E. coli — 2009 (5 R)



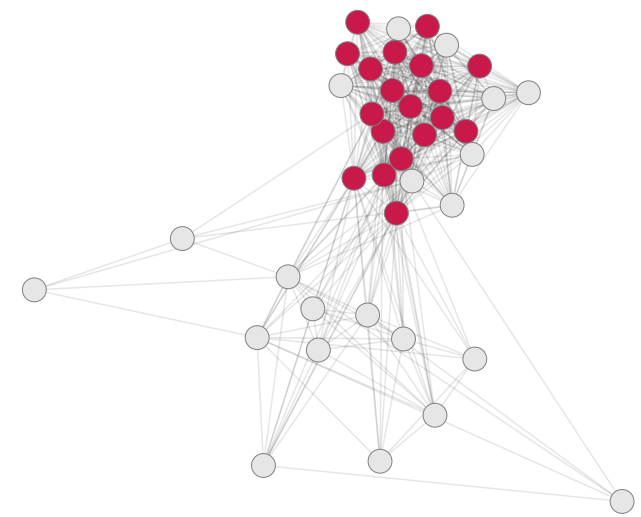
E. coli — 2014 (10 R)



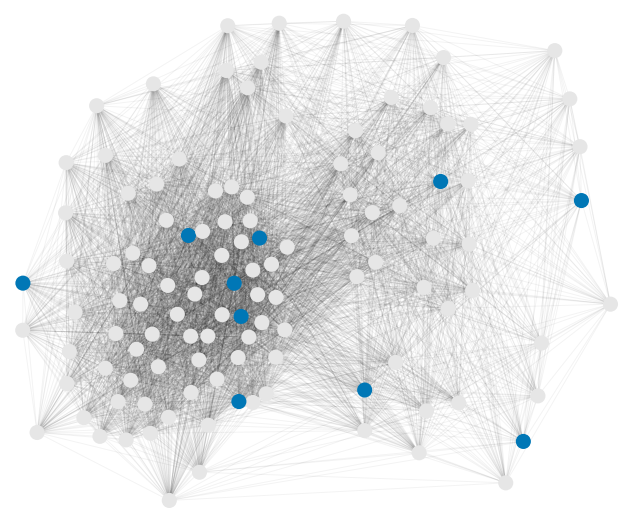
E. coli — 2019 (14 R)



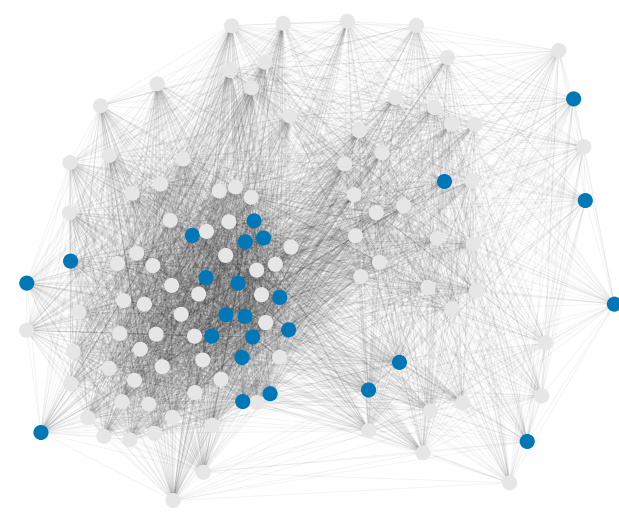
E. coli — 2024 (19 R)



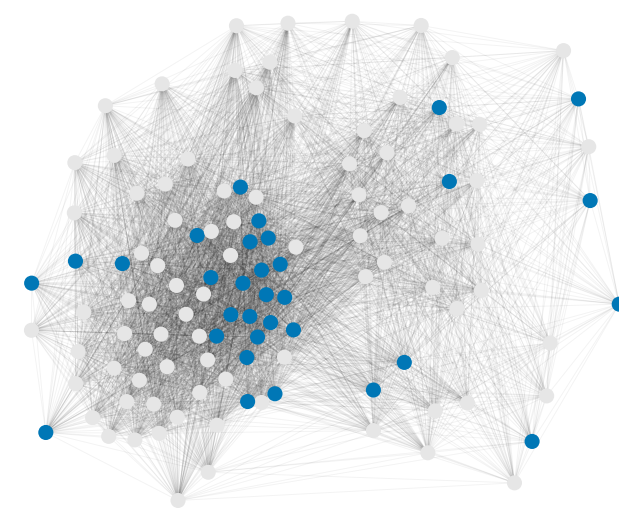
Meropenem — 2009 (10 R)



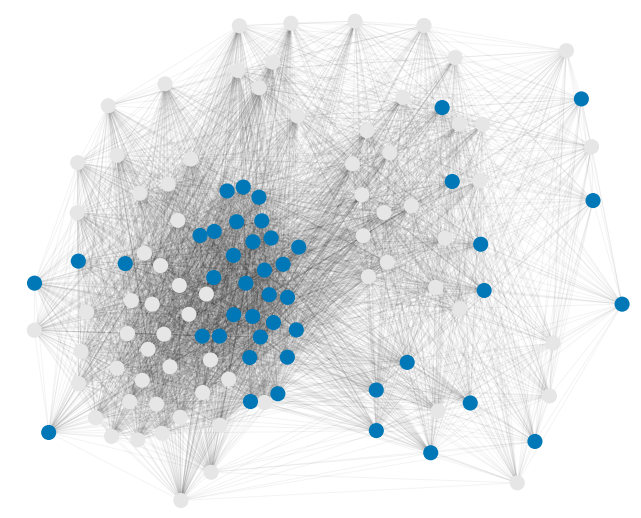
Meropenem — 2014 (25 R)



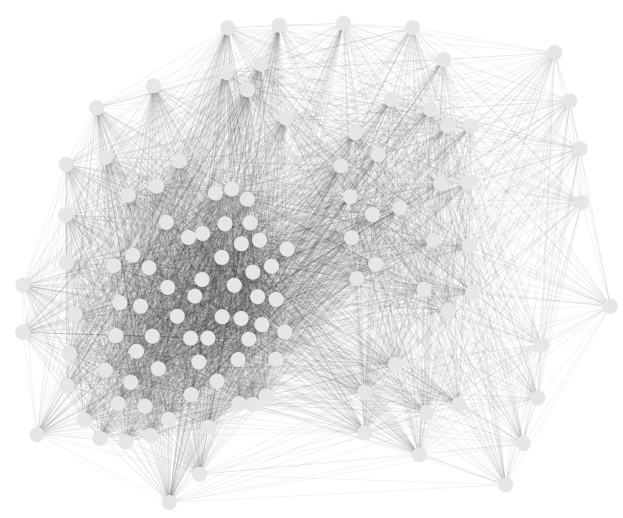
Meropenem — 2019 (32 R)



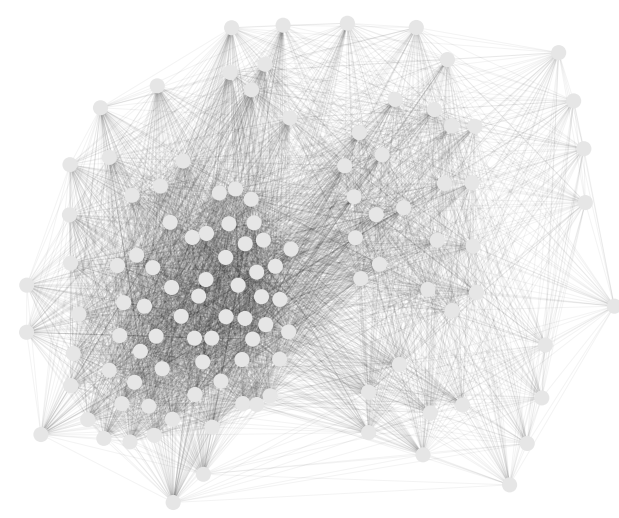
Meropenem — 2024 (45 R)



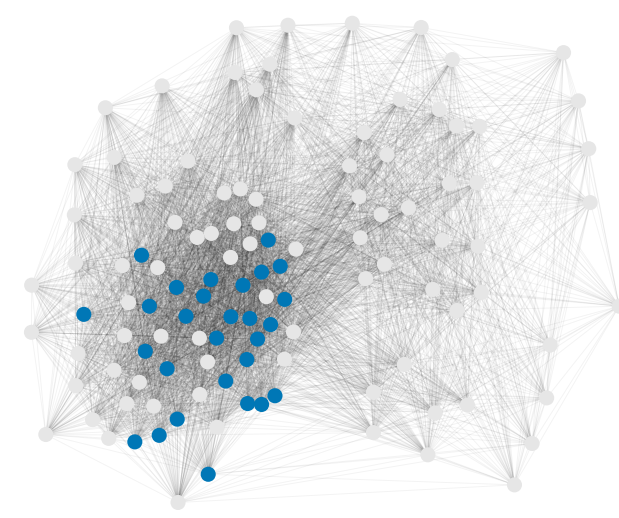
Ciprofloxacin — 2009 (0 R)



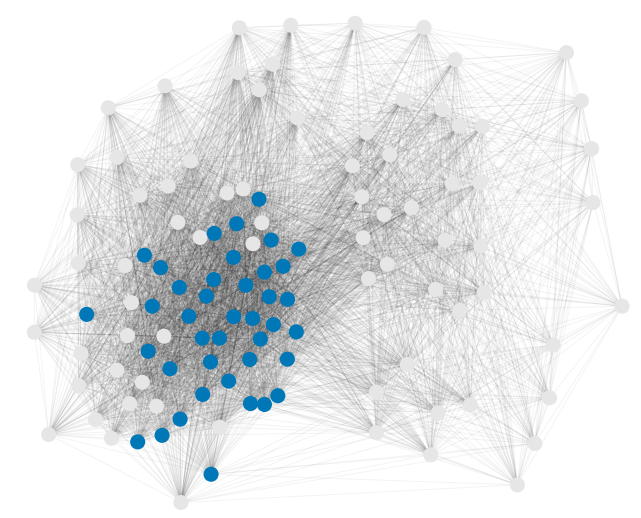
Ciprofloxacin — 2014 (0 R)



Ciprofloxacin — 2019 (28 R)



Ciprofloxacin — 2024 (40 R)



pathogen on
antibiotic network

pathogen on
antibiotic network

antibiotic on
pathogen network

antibiotic on
pathogen network